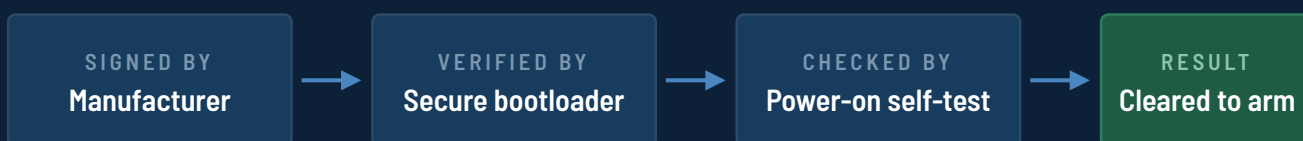


SECURE FIRMWARE FOR UNCREWED AIRCRAFT

Drone software that checks itself before every flight.

We build the security layer that lets a drone verify its own software, refuse anything we did not sign, and keep a sealed record of what happened — so operators, customers and regulators can all trust the same aircraft.



01

Verified startup on every power-on

02

Only manufacturer-signed updates install

03

Safety limits built in, not settings

04

Sealed, verifiable security log

Built to the requirements of the DGCA Certification Scheme for Unmanned Aircraft Systems, Section 7.1 (Level 1) · Validated on production flight-controller hardware

A drone is a computer that flies.

Its software decides how high it goes, how far it travels, and whether it takes off at all. Anyone who can change that software can change all of it — quietly, and without the operator noticing. **Our job is to make that impossible, and to make it provable.**

Someone alters the software on the aircraft

A modified build could lift a height limit, disable a geofence, or add behaviour nobody approved. From the outside, the drone looks completely normal.

WHAT WE DO

The aircraft carries a fingerprint of the exact software it is supposed to be running, and re-checks itself against it every time it powers on. If a single byte has changed, it will not start the software — and it will not arm.

Someone installs an update that did not come from us

Update files get copied, emailed and re-shared. A convincing file from the wrong source is one of the easiest ways into a fleet.

WHAT WE DO

Every genuine release is digitally signed in our offline signing environment. The aircraft checks that signature itself, before it installs anything. An unsigned or altered file is refused — no matter who supplies it or how it arrives.

Someone changes the safety limits from a laptop

On a standard drone, maximum height, distance and speed are just settings. Any ground station can raise them in seconds.

WHAT WE DO

On ours they are built into the software itself. An operator can fly more conservatively than the approved limit whenever a mission calls for it, but nothing on the ground can raise the limit — and every attempt is recorded.

Something goes wrong and nobody can prove what happened

After an incident, an unprotected log is only as trustworthy as the person handing it over.

WHAT WE DO

The aircraft keeps its own security record and seals it with a signature only we can produce. Anyone can be given the file; only a genuine, unaltered one will pass verification.

None of this depends on the operator remembering to do anything, and none of it can be switched off from the ground. The checks run whether or not anyone is watching.

Four checks, from our building to the flight line.

The principle is simple: the aircraft can **recognise** our signature, but it cannot **create** one. Everything else follows from that.

01 We sign the software – offline

Each release is signed on a machine that is never connected to a network. The signing key stays with us and is never shipped, never transmitted, and never stored on an aircraft or a ground station.

02 The aircraft is given the matching public half

It is built into the flight controller at manufacture. This is what lets the drone check our signature – and it is deliberately useless for forging one. Even an aircraft that fell entirely into the wrong hands could not sign anything in our name.

03 Every power-on, it verifies before it runs

A small trusted program starts first and checks the main software's signature. If the check fails, the main software is never launched. This happens on every single boot, in under a second, with nothing for the operator to do.

04 Then it checks itself again before allowing arming

Once running, the software re-calculates its own fingerprint and compares it with the registered one held on the aircraft. A mismatch blocks arming and is written to the sealed security log. The result is shown in the ground station, so the crew knows immediately why an aircraft will not fly.

THE PRACTICAL EFFECT

An aircraft running our firmware will not fly with software we did not approve. Not because someone checked a box before takeoff, but because the aircraft refuses on its own – every time, without exception, and with a record of why.

A complete package, not just a firmware file.

We work as a firmware manufacturer for drone builders taking an aircraft through type certification, and for operators who need to show their fleet is trustworthy.

Secured flight firmware

The full security layer built into the flight controller software, tuned to your aircraft's approved limits and airframe.

inoflyGCU ground station

Our ground control software. Everything a crew expects, plus clear panels for firmware status, secure updates, pairing and log download.

Signed release pipeline

Every update you ever issue is produced, fingerprinted and signed through the same controlled process – so the chain never breaks between versions.

Certification documentation

The evidence package your certification body asks for, mapped condition by condition, with test results from real hardware.

Manufacturing procedures

Written, repeatable steps for building and sealing units on a production line, including how returned aircraft are safely re-provisioned.

Hardware assessment

Bringing your own flight controller? We assess it against the requirements and tell you honestly what fits, what needs work, and what does not.

424

automated compliance checks, all passing

RSA-2048

signing with SHA-256, used throughout

<1s

added to startup by the security checks

0

network services required to stay secure

Talk to us about your aircraft

Tell us the flight controller you use and where you are in the certification process. We will tell you what is involved – before anyone commits to anything.

GET IN TOUCH

[email address]

[website]

[phone]

Inofly supplies firmware and supporting evidence as a firmware manufacturer. Type certification is granted by the relevant authority and its appointed certification body; we do not issue certificates or act as a certification body. Product names referenced remain the property of their respective owners.